Indian Institute of Technology Guwahati
Department of Physics
PH102
Tutorial-5

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1. For some spherically symmetric distribution of charges, the potential on the surface of a sphere of radius R is given to be V_0. The sphere does not contain any charge inside it. Calculate the potential in the regions $r > R$ and $r < R$.

2. A uniform line charge λ is placed on an infinite straight wire, a distance d above a grounded conducting plane. The wire runs parallel to the x-axis and directly above it, and the conducting plane is the xy plane.

   (a) Find the potential in the region above the plane.

   (b) Find the charge density a induced on the conducting plane.

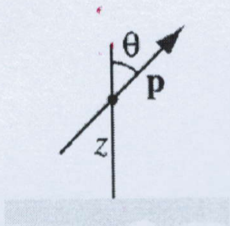
   (c) Find the total induced charge on a strip of width w parallel to the y-axis.

3. According to quantum mechanics, the electron cloud for a hydrogen atom in the ground state has a charge density

$$\rho(r) = \frac{q}{\pi a^3} e^{-2r/a}$$

   where q is the charge of the electron and a is the Bohr radius. Find the atomic polarizability of such an atom. [*Hint:* First calculate the electric field of the electron cloud, $E_e(r)$; then expand the exponential, assuming $r << a$.]

4. A dipole **p** is situated a distance z above an infinite grounded conducting plane as shown in the adjacent figure. The dipole makes an angle θ with the perpendicular to the plane. Find the torque on **p**. If the dipole is free to rotate, in what orientation will it come to rest?



5. A sphere of radius R carries a polarization $\mathbf{P}(\mathbf{r}) = k\mathbf{r}$, where k is a constant and $\mathbf{r}$ is the vector from the centre.

   (a) Calculate the bound charges σ_b and ρ_b.

   (b) Find the field inside and outside the sphere.